241801053-J B Dhejasvi -ASSIGNMENT PROBLEM

1. Example – 3 Children Assignment (Minimize Total Pay)

```
import itertools


cost = [
    [15, 10, 9],
    [9, 15, 10],
    [10, 12, 8]
]

names = ["Felicita", "Nida", "Electra"]
jobs = ["Mow", "Paint", "Wash"]

min_cost = float('inf')
best_assignment = None

for perm in itertools.permutations(range(3)):
    total = sum(cost[i][perm[i]] for i in range(3))
    if total < min_cost:
        min_cost = total
        best_assignment = perm

print("Minimum Cost:", min_cost)
for i in range(3):
    print(names[i], "->", jobs[best_assignment[i]])
```

```
Minimum Cost: 27
Felicita -> Paint
Nida -> Mow
Electra -> Wash
```

2. 4×4 Assignment Model (Minimize Cost)

```
import itertools

cost = [
    [1, 4, 6, 3],     # Felicita
    [9, 7, 10, 9],    # Nida
    [4, 5, 11, 7],    # Electra
    [8, 7, 8, 5]      # Vesper
]

names = ["Felicita", "Nida", "Electra", "Vesper"]
jobs = ["Mow", "Paint", "Wash", "Vacuum"]

min_cost = float('inf')
best_assignment = None

for perm in itertools.permutations(range(4)):
    total = sum(cost[i][perm[i]] for i in range(4))
    if total < min_cost:
        min_cost = total
        best_assignment = perm

print("Minimum Cost:", min_cost)
for i in range(4):
    print(names[i], "->", jobs[best_assignment[i]])
```

```
Minimum Cost: 21
Felicita -> Mow
Nida -> Wash
Electra -> Paint
Vesper -> Vacuum
```

3. Salespersons to Regions (Maximize Total Sales)

```
import itertools

sales = [
    [10, 22, 12, 14],  # A
    [16, 18, 22, 10],  # B
    [24, 20, 12, 18],  # C
    [16, 14, 24, 20]   # D
]

names = ["A", "B", "C", "D"]
regions = ["I", "II", "III", "IV"]

max_sales = 0
best_assignment = None

for perm in itertools.permutations(range(4)):
    total = sum(sales[i][perm[i]] for i in range(4))
    if total > max_sales:
        max_sales = total
        best_assignment = perm

print("Maximum Sales:", max_sales)
for i in range(4):
    print(names[i], "-> Region", regions[best_assignment[i]])
```

```
Maximum Sales: 88
A -> Region II
B -> Region III
C -> Region I
D -> Region IV
```

4. Travelling Salesman Problem (Minimum Tour Time)

```
import itertools

# Distance matrix
dist = [
    [0, 5, 8, 4, 5],
    [5, 0, 7, 4, 5],
    [8, 7, 0, 8, 6],
    [4, 4, 8, 0, 8],
    [5, 5, 6, 8, 0]
]

cities = ["I", "II", "III", "IV", "V"]

min_cost = float('inf')
best_path = None

# Fix starting city as 0 (City I)
for perm in itertools.permutations(range(1, 5)):
    path = (0,) + perm + (0,)
    total = sum(dist[path[i]][path[i+1]] for i in range(len(path)-1))

    if total < min_cost:
        min_cost = total
        best_path = path

print("Minimum Travel Time:", min_cost)
print("Tour:")
for city in best_path:
    print(cities[city], end=" -> ")
```

```
Minimum Travel Time: 26
Tour:
I -> IV -> II -> III -> V -> I ->
```